Cyclohexane Reactor Ver 1 · operator training simulator

Heat and material balance

Document no. CHX-V1-PE-HMB-001 · **Discipline** Process engineering

Client Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University (teaching)

- Three cases from the simulator model: **Design** (benzene 34.8 t/h), **Off-design 1 · 120 %** (41.76 t/h, hydrogen in ratio) and **Off-design 2 · 70 %** (24.36 t/h).
- 25 streams per case: the 23 streams of PFD CHX-V1-PFD-001 plus the FV-104 by-pass (10B) and the D-101 blowdown (BD).
- Overall mass balance closure: design −0.0066 %, 120 % −0.0036 %, 70 % 0.0066 %; every unit within 0.0070 %.
- Product 37.50 / 45.00 / 26.24 t/h with 2.00 / 6.54 / 0.20 ppm benzene; LP steam 14.72 / 18.02 / 9.60 t/h.
- Off-design cases: steady states of the Operating-windows engine (owRun, 4 h of plant time); design case = the model’s design state (noise off).

Revisions

Rev	Date	Description	Originator	Checker	Approver
0	2026-10-26	Issued for teaching	Apinan Soottitantawat		

This document is generated for teaching from the simulator model of the unit. It is not issued for construction; every number comes from the model or is labelled assumed / typical practice. No client, contractor or vendor is implied.

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1 · Design basis and cases

Design basis (the model's design card and its stated assumptions): 300 kt/y cyclohexane at 8000 h/y → benzene 34.8 t/h; $C_6H_6 + 3\ H_2 \rightarrow C_6H_{12}$, $\Delta H = -206$ kJ/mol; make-up hydrogen 99.5 % H_2, 0.5 % CH_4; cooling water 32 °C; MP steam 10 bar g; LP steam header 3.5 bar g; BFW 105 °C, 10 bar g; nitrogen 8 bar g. Molar masses as in the model: benzene 78.11, cyclohexane 84.16, H_2 2.016, CH_4 16.04, water 18.015.

Case	Definition	Why this case
Design case (normal operation, 100 % rate)	The simulator at its design point: benzene 34.8 t/h, FIC-103 46 000 Nm³/h, every loop in its normal mode at its design set point, CW 32 °C, HIC-107 25 %, fresh catalyst. Same state as the page's snapDesign (noise off).	The normal operating point: the reference for the PFD flags, the equipment process datasheets and the normal-flow column of the line list.
Off-design case 1 · 120 % rate, hydrogen in ratio (rated case)	Benzene 41.76 t/h (120 %), FIC-103 set in ratio to the feed (55 200 Nm³/h), all other set points at design, CW 32 °C, HIC-107 25 %, fresh catalyst. owRun to steady state (4 h of plant time).	The design-basis maximum rate. It sits just inside the top of the operating window (42.0 t/h, Operating windows tab), so it fixes the rated flows of lines, P-101, K-101 and the maximum openings of the control valves.
Off-design case 2 · 70 % turndown, hydrogen in ratio	Benzene 24.36 t/h (70 %), FIC-103 in ratio (32 200 Nm³/h), all other set points at design. owRun to steady state (4 h of plant time).	The minimum continuous case: lowest velocities in the catalyst-slurry lines, minimum openings of FV-103, FV-104, PV-110 and the level valves, least LP steam. 70 % is well above the bottom of the window (7.8 t/h), so the case is a realistic turndown, not a limit.

Considered and not chosen: End-of-run R-101 catalyst (activity 0.6, design rate). At the design rate it moves product benzene 2.0 → 5.2 ppm and R-101 conversion to 94.8 %, and lowers the loads (pump-around 357 → 304 t/h, LP steam 12.8 t/h); no battery-limit stream changes by more than 0.1 %. It sets no line size and no rated duty, so a rate case was preferred.

Key numbers per case

Key number	Design	Off-design 1 · 120 %	Off-design 2 · 70 %
Benzene feed (FIC-101) t/h	34.80	41.76	24.36
Hydrogen to R-101 (FIC-103) Nm³/h	46,000	55,200	32,200
Make-up hydrogen (FI-106) Nm³/h	31,974	37,966	22,987
Purge (FI-107) Nm³/h	2,017	2,017	2,017
Recycle purity (AI-105) %	92.07	90.59	94.30
R-101 temperature °C	200.0	200.0	200.0
R-101 pressure bar g	30.00	30.44	29.49
R-101 conversion %	98.00	97.14	98.89
R-102 temperature °C	160.0	160.0	160.0
R-103 outlet temperature °C	158.6	158.6	158.5
Product (FI-110) t/h	37.50	45.00	26.24
Product benzene ppm	2.00	6.54	0.20
Product MCP ppm	60	60	60
Pump-around through E-101 t/h	357	454	224
Pump-around return °C	159.3	161.0	157.1
E-101 duty kW	9,598	11,691	6,349
LP steam t/h	14.72	18.02	9.60

Key number	Design	Off-design 1 · 120 %	Off-design 2 · 70 %
Reflux t/h	46.5	55.2	33.1
TI-103 °C	55.7	58.7	51.1
V-104 pressure bar g	29.00	29.00	29.00
K-101 discharge bar g	32.00	31.29	32.82
FV-103 opening %	50.0	92.1	27.1
FV-101 opening %	61.9	74.3	43.3
PV-105 opening %	50.0	59.4	35.9
FV-104 opening %	45.4	57.6	28.5
PV-110 opening %	50.0	61.2	32.6
TV-102 opening %	50.0	42.9	51.1
H_2 consumed Nm^3/h	29,958	35,949	20,971
Make-up H_2 (as H_2) Nm^3/h	31,815	37,776	22,872
H_2 purged Nm^3/h	1,857	1,827	1,902
Margin, econ() training prices $/h	7,466	9,051	5,096

2 · Stream tables

Composition in mol % of the whole stream (ppm where below 0.1 %). MCP is the model's isomer term in ppm (a trace, not carried in the mass balance). Phase: L liquid, V vapour, V/L two-phase. VF = molar vapour fraction. Density and volumetric flow are estimates: liquid from the model's molar density (benzene from a typical correlation), gas as ideal gas (steam Z 0.95), two-phase homogeneous. Streams 16–17: phase split by a flash estimate with the model's own PsatC (the model lumps R-103 vapour with the liquid). Stream 1 pressure assumed (1.0 bar g). 10B and BD are not numbered on the PFD (EQ-ep-1).

Design case (normal operation, 100 % rate) — streams 1 to 9

Stream no.	1	2	3	4	5	6	7	8	9
Stream	Benzene from TK-100	Benzene feed to R-101	Recycle + make-up H_2 to R-101 sparger	R-101 overhead (gas + vapour)	E-103 outlet	Reflux to R-101	Gas to R-102 sparger	R-101 slurry to R-102 (LV-101)	Pump-around to P-101
From → to	TK-100 → P-100A/B	P-100A/B (FV-101, XV-101) → R-101	K-101 (FV-103) → R-101	R-101 → E-103	E-103 → V-103	V-103 → R-101	V-103 → R-102	R-101 → R-102	R-101 → P-101A/B
Phase	L	L	V	V	V/L	L	V	L	L
Temperature °C	40.0	40.0	60.0	200.0	55.7	55.7	55.7	200.0	200.0
Pressure bar g	1.00*	30.00	32.00	30.00	30.00	30.00	29.60	30.00	30.00
Mass flow t/h	34.80	34.80	6.419	51.19	51.19	46.46	4.727	36.49	605.08
Molar flow kmol/h	445.5	445.5	2,052.3	1,306.7	1,306.7	552.9	753.8	434.2	7,200.0
Vapour fraction (mol)	0.000	0.000	1.000	1.000	0.577	0.000	1.000	0.000	0.000
H_2 mol %	—	—	92.07	44.37	44.37	—	76.92	—	—
CH_4 mol %	—	—	7.93	12.45	12.45	—	21.59	—	—
Benzene mol %	100.00	100.00	—	0.91	0.91	2.10	314 ppm	2.00	2.00
Cyclohexane mol %	—	—	—	42.27	42.27	97.90	1.47	98.00	98.00
Water mol %	—	—	—	—	—	—	—	—	—
MCP ppm	—	—	—	—	—	—	—	45	45
Molar mass kg/kmol	78.11	78.11	3.13	39.18	39.18	84.03	6.27	84.04	84.04
Density kg/m³ (est.)	858	858	3.73	30.9	70.3	735	7.02	607	607
Actual flow m³/h (est.)	40.6	40.6	1,722	1,658	728	63.2	673	60.1	996

Design case (normal operation, 100 % rate) — streams 10 to 17

Stream no.	10	10B	11	12	13	14	15	16	17
Stream	Pump-around to E-101 (FV-104)	FV-104 by-pass of E-101 (not numbered on the PFD)	Pump-around return from E-101	R-102 gas + vapour to R-103	R-102 slurry to P-102	P-102 discharge to F-102	Filtered liquid to R-103 (LV-102)	R-103 effluent	E-106 outlet
From → to	P-101A/B → E-101	FV-104 → R-101	E-101 → R-101	R-102 → R-103	R-102 → P-102A/B	P-102A/B → F-102A/B	F-102A/B → R-103	R-103 → E-106	E-106 → V-104
Phase	L	L	L	V	L	L	L	V/L	V/L
Temperature °C	200.0	200.0	159.3	160.0	160.0	160.0	160.0	158.6	44.9
Pressure bar g	32.00	30.00	30.00	29.60	29.60	32.60	29.25	29.25	29.00
Mass flow t/h	356.79	248.29	356.79	20.28	20.94	20.94	20.94	41.22	41.22
Molar flow kmol/h	4,245.5	2,954.5	4,245.5	912.6	248.8	248.8	248.8	1,161.3	1,161.3
Vapour fraction (mol)	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.782	0.623
H_2 mol %	—	—	—	60.61	—	—	—	47.62	47.62
CH_4 mol %	—	—	—	17.83	—	—	—	14.01	14.01
Benzene mol %	2.00	2.00	2.00	23 ppm	100 ppm	100 ppm	100 ppm	0.8 ppm	0.8 ppm
Cyclohexane mol %	98.00	98.00	98.00	21.56	99.99	99.99	99.99	38.37	38.37
Water mol %	—	—	—	—	—	—	—	—	—
MCP ppm	45	45	45	—	45	45	45	60	60
Molar mass kg/kmol	84.04	84.04	84.04	22.23	84.16	84.16	84.16	35.50	35.50
Density kg/m³ (est.)	607	607	643	18.9	641	641	641	37.1	60.0
Actual flow m³/h (est.)	588	409	555	1,074	32.6	32.6	32.6	1,111	687

Design case (normal operation, 100 % rate) — streams 18 to BD

Stream no.	18	19	20	21	22	23	BD		
Stream	Recycle gas to K-101	Make-up hydrogen (PV-105)	Purge to fuel gas (HIC-107)	Cyclohexane product (LV-104)	LP steam export (PV-110)	Boiler feed water (LV-109)	D-101 blowdown (not numbered on the PFD)		
From → to	V-104 → K-101	battery limit → V-104	V-104 → fuel gas	V-104 → TK-106 / TK-107	D-101 → LP steam header	BFW header → D-101	D-101 → blowdown		
Phase	V	V	V	L	V	L	L		
Temperature °C	44.9	40.0	44.9	44.9	155.8	105.0	155.8		
Pressure bar g	29.00	34.99	29.00	29.00	4.50	9.99	4.50		
Mass flow t/h	6.419	2.976	0.281	37.50	14.72	14.87	0.147		
Molar flow kmol/h	2,052.3	1,426.5	90.0	445.5	817.3	825.4	8.2		
Vapour fraction (mol)	1.000	1.000	1.000	0.000	1.000	0.000	0.000		
H_2 mol %	92.07	99.50	92.07	—	—	—	—		
CH_4 mol %	7.93	0.50	7.93	—	—	—	—		
Benzene mol %	—	—	—	2 ppm	—	—	—		
Cyclohexane mol %	—	—	—	100.00	—	—	—		
Water mol %	—	—	—	—	100.00	100.00	100.00		
MCP ppm	—	—	—	60	—	—	—		
Molar mass kg/kmol	3.13	2.09	3.13	84.16	18.02	18.02	18.02		
Density kg/m³ (est.)	3.55	2.88	3.55	743	2.93	954	903		
Actual flow m³/h (est.)	1,808	1,032	79.3	50.5	5,023	15.6	0.163		

Off-design case 1 · 120 % rate, hydrogen in ratio (rated case) — streams 1 to 9

Stream no.	1	2	3	4	5	6	7	8	9
Stream	Benzene from TK-100	Benzene feed to R-101	Recycle + make-up H_2 to R-101 sparger	R-101 overhead (gas + vapour)	E-103 outlet	Reflux to R-101	Gas to R-102 sparger	R-101 slurry to R-102 (LV-101)	Pump-around to P-101
From → to	TK-100 → P-100A/B	P-100A/B (FV-101, XV-101) → R-101	K-101 (FV-103) → R-101	R-101 → E-103	E-103 → V-103	V-103 → R-101	V-103 → R-102	R-101 → R-102	R-101 → P-101A/B
Phase	L	L	V	V	V/L	L	V	L	L
Temperature °C	40.0	40.0	60.0	200.0	58.7	58.7	58.7	200.0	200.0
Pressure bar g	1.00*	30.44	31.29	30.44	30.44	30.44	29.86	30.44	30.44
Mass flow t/h	41.76	41.76	8.216	61.40	61.40	55.07	6.333	43.64	604.71
Molar flow kmol/h	534.6	534.6	2,462.8	1,575.4	1,575.4	655.7	919.7	519.7	7,200.0
Vapour fraction (mol)	0.000	0.000	1.000	1.000	0.584	0.000	1.000	0.000	0.000
H_2 mol %	—	—	90.59	42.71	42.71	—	73.16	—	—
CH_4 mol %	—	—	9.41	14.72	14.72	—	25.21	—	—
Benzene mol %	100.00	100.00	—	1.28	1.28	3.00	488 ppm	2.85	2.85
Cyclohexane mol %	—	—	—	41.30	41.30	97.00	1.58	97.15	97.15
Water mol %	—	—	—	—	—	—	—	—	—
MCP ppm	—	—	—	—	—	—	—	45	45
Molar mass kg/kmol	78.11	78.11	3.34	38.97	38.97	83.98	6.89	83.99	83.99
Density kg/m³ (est.)	858	858	3.89	31.2	69.6	733	7.71	608	608
Actual flow m³/h (est.)	48.7	48.7	2,112	1,970	882	75.1	822	71.8	995

Off-design case 1 · 120 % rate, hydrogen in ratio (rated case) — streams 10 to 17

Stream no.	10	10B	11	12	13	14	15	16	17
Stream	Pump-around to E-101 (FV-104)	FV-104 by-pass of E-101 (not numbered on the PFD)	Pump-around return from E-101	R-102 gas + vapour to R-103	R-102 slurry to P-102	P-102 discharge to F-102	Filtered liquid to R-103 (LV-102)	R-103 effluent	E-106 outlet
From → to	P-101A/B → E-101	FV-104 → R-101	E-101 → R-101	R-102 → R-103	R-102 → P-102A/B	P-102A/B → F-102A/B	F-102A/B → R-103	R-103 → E-106	E-106 → V-104
Phase	L	L	L	V	L	L	L	V/L	V/L
Temperature °C	200.0	200.0	161.0	160.0	160.0	160.0	160.0	158.6	44.9
Pressure bar g	32.44	30.44	30.44	29.86	29.86	32.86	29.36	29.36	29.00
Mass flow t/h	452.85	151.86	452.85	24.64	25.34	25.34	25.34	49.98	49.98
Molar flow kmol/h	5,391.9	1,808.1	5,391.9	1,092.7	301.1	301.1	301.1	1,393.5	1,393.5
Vapour fraction (mol)	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.782	0.623
H_2 mol %	—	—	—	57.41	—	—	—	45.00	45.00
CH_4 mol %	—	—	—	21.22	—	—	—	16.64	16.64
Benzene mol %	2.85	2.85	2.85	40 ppm	179 ppm	179 ppm	179 ppm	3 ppm	3 ppm
Cyclohexane mol %	97.15	97.15	97.15	21.37	99.98	99.98	99.98	38.37	38.37
Water mol %	—	—	—	—	—	—	—	—	—
MCP ppm	45	45	45	—	45	45	45	60	60
Molar mass kg/kmol	83.99	83.99	83.99	22.55	84.16	84.16	84.16	35.86	35.86
Density kg/m³ (est.)	608	608	642	19.3	641	641	641	37.6	60.6
Actual flow m³/h (est.)	745	250	705	1,275	39.5	39.5	39.5	1,328	824

Off-design case 1 · 120 % rate, hydrogen in ratio (rated case) — streams 18 to BD

Stream no.	18	19	20	21	22	23	BD		
Stream	Recycle gas to K-101	Make-up hydrogen (PV-105)	Purge to fuel gas (HIC-107)	Cyclohexane product (LV-104)	LP steam export (PV-110)	Boiler feed water (LV-109)	D-101 blowdown (not numbered on the PFD)		
From → to	V-104 → K-101	battery limit → V-104	V-104 → fuel gas	V-104 → TK-106 / TK-107	D-101 → LP steam header	BFW header → D-101	D-101 → blowdown		
Phase	V	V	V	L	V	L	L		
Temperature °C	44.9	40.0	44.9	44.9	155.8	105.0	155.8		
Pressure bar g	29.00	34.99	29.00	29.00	4.50	9.99	4.50		
Mass flow t/h	8.216	3.534	0.300	45.00	18.02	18.20	0.180		
Molar flow kmol/h	2,462.8	1,693.9	90.0	534.6	1,000.3	1,010.3	10.0		
Vapour fraction (mol)	1.000	1.000	1.000	0.000	1.000	0.000	0.000		
H_2 mol %	90.59	99.50	90.59	—	—	—	—		
CH_4 mol %	9.41	0.50	9.41	—	—	—	—		
Benzene mol %	—	—	—	7 ppm	—	—	—		
Cyclohexane mol %	—	—	—	100.00	—	—	—		
Water mol %	—	—	—	—	100.00	100.00	100.00		
MCP ppm	—	—	—	60	—	—	—		
Molar mass kg/kmol	3.34	2.09	3.34	84.16	18.02	18.02	18.02		
Density kg/m³ (est.)	3.79	2.88	3.79	743	2.93	954	903		
Actual flow m³/h (est.)	2,170	1,225	79.3	60.6	6,148	19.1	0.200		

Off-design case 2 · 70 % turndown, hydrogen in ratio — streams 1 to 9

Stream no.	1	2	3	4	5	6	7	8	9
Stream	Benzene from TK-100	Benzene feed to R-101	Recycle + make-up H_2 to R-101 sparger	R-101 overhead (gas + vapour)	E-103 outlet	Reflux to R-101	Gas to R-102 sparger	R-101 slurry to R-102 (LV-101)	Pump-around to P-101
From → to	TK-100 → P-100A/B	P-100A/B (FV-101, XV-101) → R-101	K-101 (FV-103) → R-101	R-101 → E-103	E-103 → V-103	V-103 → R-101	V-103 → R-102	R-101 → R-102	R-101 → P-101A/B
Phase	L	L	V	V	V/L	L	V	L	L
Temperature °C	40.0	40.0	60.0	200.0	51.1	51.1	51.1	200.0	200.0
Pressure bar g	1.00*	29.49	32.82	29.49	29.49	29.49	29.29	29.49	29.49
Mass flow t/h	24.36	24.36	4.044	35.83	35.83	33.09	2.747	25.66	605.47
Molar flow kmol/h	311.9	311.9	1,436.6	911.6	911.6	393.5	518.2	305.1	7,200.0
Vapour fraction (mol)	0.000	0.000	1.000	1.000	0.568	0.000	1.000	0.000	0.000
H_2 mol %	—	—	94.30	47.12	47.12	—	82.90	—	—
CH_4 mol %	—	—	5.70	8.98	8.98	—	15.80	—	—
Benzene mol %	100.00	100.00	—	0.51	0.51	1.17	152 ppm	1.11	1.11
Cyclohexane mol %	—	—	—	43.39	43.39	98.83	1.29	98.89	98.89
Water mol %	—	—	—	—	—	—	—	—	—
MCP ppm	—	—	—	—	—	—	—	45	45
Molar mass kg/kmol	78.11	78.11	2.82	39.31	39.31	84.09	5.30	84.09	84.09
Density kg/m³ (est.)	858	858	3.44	30.5	71.3	739	5.96	607	607
Actual flow m³/h (est.)	28.4	28.4	1,176	1,176	503	44.8	461	42.3	998

Off-design case 2 · 70 % turndown, hydrogen in ratio — streams 10 to 17

Stream no.	10	10B	11	12	13	14	15	16	17
Stream	Pump-around to E-101 (FV-104)	FV-104 by-pass of E-101 (not numbered on the PFD)	Pump-around return from E-101	R-102 gas + vapour to R-103	R-102 slurry to P-102	P-102 discharge to F-102	Filtered liquid to R-103 (LV-102)	R-103 effluent	E-106 outlet
From → to	P-101A/B → E-101	FV-104 → R-101	E-101 → R-101	R-102 → R-103	R-102 → P-102A/B	P-102A/B → F-102A/B	F-102A/B → R-103	R-103 → E-106	E-106 → V-104
Phase	L	L	L	V	L	L	L	V/L	V/L
Temperature °C	200.0	200.0	157.1	160.0	160.0	160.0	160.0	158.5	44.9
Pressure bar g	31.49	29.49	29.49	29.29	29.29	32.29	29.12	29.12	29.00
Mass flow t/h	224.21	381.25	224.21	13.90	14.51	14.51	14.51	28.40	28.40
Molar flow kmol/h	2,666.3	4,533.7	2,666.3	640.5	172.4	172.4	172.4	812.9	812.9
Vapour fraction (mol)	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.783	0.623
H_2 mol %	—	—	—	65.44	—	—	—	51.56	51.56
CH_4 mol %	—	—	—	12.78	—	—	—	10.07	10.07
Benzene mol %	1.11	1.11	1.11	8 ppm	36 ppm	36 ppm	36 ppm	0.1 ppm	0.1 ppm
Cyclohexane mol %	98.89	98.89	98.89	21.78	100.00	100.00	100.00	38.37	38.37
Water mol %	—	—	—	—	—	—	—	—	—
MCP ppm	45	45	45	—	45	45	45	60	60
Molar mass kg/kmol	84.09	84.09	84.09	21.70	84.16	84.16	84.16	34.94	34.94
Density kg/m³ (est.)	607	607	645	18.3	641	641	641	36.4	59.1
Actual flow m³/h (est.)	370	628	348	761	22.6	22.6	22.6	781	481

Off-design case 2 · 70 % turndown, hydrogen in ratio — streams 18 to BD

Stream no.	18	19	20	21	22	23	BD		
Stream	Recycle gas to K-101	Make-up hydrogen (PV-105)	Purge to fuel gas (HIC-107)	Cyclohexane product (LV-104)	LP steam export (PV-110)	Boiler feed water (LV-109)	D-101 blowdown (not numbered on the PFD)		
From → to	V-104 → K-101	battery limit → V-104	V-104 → fuel gas	V-104 → TK-106 / TK-107	D-101 → LP steam header	BFW header → D-101	D-101 → blowdown		
Phase	V	V	V	L	V	L	L		
Temperature °C	44.9	40.0	44.9	44.9	155.8	105.0	155.8		
Pressure bar g	29.00	34.99	29.00	29.00	4.50	9.99	4.50		
Mass flow t/h	4.044	2.139	0.253	26.24	9.604	9.700	0.096		
Molar flow kmol/h	1,436.6	1,025.6	90.0	311.8	533.1	538.4	5.3		
Vapour fraction (mol)	1.000	1.000	1.000	0.000	1.000	0.000	0.000		
H_2 mol %	94.30	99.50	94.30	—	—	—	—		
CH_4 mol %	5.70	0.50	5.70	—	—	—	—		
Benzene mol %	—	—	—	0.2 ppm	—	—	—		
Cyclohexane mol %	—	—	—	100.00	—	—	—		
Water mol %	—	—	—	—	100.00	100.00	100.00		
MCP ppm	—	—	—	60	—	—	—		
Molar mass kg/kmol	2.82	2.09	2.82	84.16	18.02	18.02	18.02		
Density kg/m³ (est.)	3.20	2.88	3.20	743	2.93	954	903		
Actual flow m³/h (est.)	1,266	742	79.3	35.3	3,277	10.2	0.106		

3 · Mass balance per unit (in = out)

Unit	In (streams)	Out (streams)	Design in t/h	out t/h	in − out %	Off-design 1 · 120 % in t/h	out t/h	in − out %	Off-design 2 · 70 % in t/h	out t/h	in − out %
P-100A/B	1	2	34.800	34.800	0.0000	41.760	41.760	0.0000	24.360	24.360	0.0000
R-101 (with sparger, reflux, pump-around nozzles)	2 + 3 + 6 + 10B + 11	4 + 8 + 9	692.764	692.765	−0.0001	709.753	709.754	−0.0001	666.958	666.958	−0.0001
P-101A/B + FV-104	9	10 + 10B	605.081	605.081	0.0000	604.710	604.710	0.0000	605.467	605.467	0.0000
E-101 (process side)	10	11	356.789	356.789	0.0000	452.848	452.848	0.0000	224.215	224.215	0.0000
E-103	4	5	51.191	51.191	0.0000	61.400	61.400	0.0000	35.833	35.833	0.0000
V-103	5	6 + 7	51.191	51.191	0.0000	61.400	61.400	0.0000	35.833	35.833	0.0000
R-102	7 + 8	12 + 13	41.220	41.220	−0.0009	49.977	49.977	−0.0001	28.405	28.405	0.0008
P-102A/B	13	14	20.937	20.937	0.0000	25.337	25.337	0.0000	14.507	14.507	0.0000
F-102 + LV-102	14	15	20.937	20.937	0.0000	25.337	25.337	0.0000	14.507	14.507	0.0000
R-103	12 + 15	16	41.220	41.220	0.0000	49.977	49.977	0.0000	28.405	28.405	0.0000
E-106	16	17	41.220	41.220	0.0000	49.977	49.977	0.0000	28.405	28.405	0.0000
V-104	17 + 19	18 + 20 + 21	44.196	44.198	−0.0028	53.511	53.511	−0.0010	30.544	30.542	0.0070
K-101	18	3	6.419	6.419	0.0000	8.216	8.216	0.0000	4.044	4.044	0.0000
D-101 + E-101 water side	23	22 + BD	14.869	14.870	−0.0069	18.200	18.200	0.0003	9.700	9.700	−0.0017
Overall (process battery limits)	1 + 19	20 + 21	37.776	37.778	−0.0066	45.294	45.295	−0.0036	26.499	26.498	0.0066

Every boundary closes within 0.01 %. The small residuals come from the molar masses of the model (78.11 + 3 × 2.016 = 84.158 vs 84.16 for cyclohexane) and from the inventories still settling at the end of each run.

Hydrogen balance (Nm³/h of H_2)

Case	Make-up H_2 in	Consumed (3 × reaction)	Purged	In − out %
Design	31,815	29,958	1,857	0.000
Off-design 1 · 120 %	37,776	35,949	1,827	0.000
Off-design 2 · 70 %	22,872	20,971	1,902	0.000

Overall mass balance closure

Case	Benzene in t/h	Make-up gas in t/h	Product out t/h	Purge out t/h	In − out t/h	Closure %
Design	34.800	2.976	37.497	0.281	−0.0025	−0.0066
Off-design 1 · 120 %	41.760	3.534	44.995	0.300	−0.0016	−0.0036
Off-design 2 · 70 %	24.360	2.139	26.244	0.253	0.0017	0.0066

4 · Duties and utility summary

Duty kW (+ = released or added)	Design	Off-design 1 · 120 %	Off-design 2 · 70 %
R-101 heat of reaction	24,983	29,719	17,647
R-102 heat of reaction	508	868	198
R-103 heat of reaction	3	5	1
E-101 pump-around cooler → LP steam (D-101)	9,598	11,691	6,349
E-103 reflux condenser (CW)	7,872	9,358	5,584
E-106 product cooler (CW), sensible estimate	3,492	4,192	2,443
R-102 jacket (+ heating with MP steam)	1,293	1,108	1,321
M-101 agitator power (model formula)	153	153	153

Utility	Design	Off-design 1 · 120 %	Off-design 2 · 70 %
LP steam raised (FI-111) t/h	14.72	18.02	9.60
MP steam to R-102 jacket (FI-108) t/h	2.33	2.00	2.38
MP steam to E-101 (HS-101 HEAT only) t/h	0.00	0.00	0.00
Boiler feed water (FI-109) t/h	14.87	18.20	9.70
D-101 blowdown (1 % of steam) t/h	0.147	0.180	0.096
Cooling water to E-103 t/h (model K.Fcw3)	432	432	432
Cooling water to E-106 t/h (econ() term, assumed)	216	216	216
Cooling water to R-102 jacket (COOL only) t/h	0	0	0
CW return E-103 °C	47.7	50.7	43.1
CW return E-106 °C	45.9	48.7	41.7
Power K-101 kW (econ())	1,200	1,440	840
Power M-101 kW (JI-101)	153	153	153
Power P-100 kW	90	90	90
Power P-101 kW	160	160	160
Power P-102 kW	30	30	30
Power total kW	1,633	1,873	1,273
Nitrogen, normal operation Nm³/h	0	0	0
Fuel gas export (purge) Nm³/h	2,017	2,017	2,017
Fuel gas export kg/h	281	300	253

Power figures are the econ() loads of the model (training values); Process engineering part 6 compares them with flow × head estimates (EQ-ep-10). The model has no cooling-water flow for E-106; the 60 kg/s is the fixed CW term of econ(). Nitrogen is used only for purging (HV-120).

5 · Stream numbering, notes and holds

PFD stream numbers (drawing CHX-V1-PFD-001) and the Process-tab (BFD) stream numbers of the same streams:

PFD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
BFD	—	1	2	3	—	4	5	6	14	14	14	7	8	8	8	9	—	10	11	12	13	15	16

Holds and engineering queries affecting this document

ID	Query	Finding	Proposed resolution
EQ-ep-1	PFD has no stream for the FV-104 by-pass and the D-101 blowdown	P-101 always moves 606 t/h; at design 248 t/h of it by-passes E-101 through the three-way FV-104 (stream 10B here), and D-101 blows down 1 % of the steam (BD). PFD stream 10 shows only the 357 t/h through E-101, so the PFD alone under-states the pump and the return line by 70 %.	Add 10B and BD to PFD rev 1; P-101 and lines 16"-P-1004-B3, 16"-P-1006-B3 rated at the full pump flow.
EQ-ep-7	Cooling-water return temperatures above typical practice	E-103 returns 47.7 °C at K.Fcw3 = 120 kg/s and E-106 45.9 °C at the econ() 60 kg/s (design); at 120 % E-106 returns 48.7 °C. Typical maximum return 45 °C (fouling, scaling). The model has no E-106 water flow.	E-103 CW ≥ 145 kg/s and E-106 ≥ 77 kg/s for a 45 °C return; put the E-106 flow on its datasheet.
EQ-ep-8	Recycle gas carries no cyclohexane vapour in the model	At V-104 (44.9 °C, 30.0 bar a) the model's own PsatC gives 1.1 mol % cyclohexane in the gas ≈ 1.84 t/h circulating through K-101 and 81 kg/h lost to fuel gas; the HMB follows the model (dry gas).	K-101 datasheet with the saturated gas (molar mass and flow); product loss to fuel gas in the economics.
EQ-ep-9	FV-103 at 92 % open in the 120 % case	The design-basis maximum rate needs FV-103 at 92.1 % (design 50 %, 70 % case 27 %); typical practice keeps a control valve ≤ 80–90 % open at maximum flow. The Operating-windows tab finds FV-103 as the top of the window.	Size FV-103 for the 120 % case at ≤ 80 % (larger Cv) together with K-101 head (Flexibility recommendation).
EQ-ep-10	Power in econ() differs from the hydraulic estimates	econ() books K-101 1200 kW, P-101 160 kW, P-100 90 kW, P-102 30 kW; part 6 estimates the absorbed power from flow × head (see the table).	Electrical load list from the equipment datasheets (absorbed power + motor margin); keep econ() as a training price model and say so.

Assumptions

Feed treated as pure benzene (the model has no impurity; design basis 99.9 %). Gas inerts counted as CH_4 (the model's inert inventory). Liquid viscosity and density correlations, ideal-gas densities and the flash estimate of streams 16–17 are labelled estimates. Off-design cases: owRun with the trips moved out of reach inside the run only (none came near a trip), CW 32 °C, purge HIC-107 25 %, fresh catalyst, set points at design, hydrogen in ratio to the feed. Stream 1 pressure 1 bar g (assumed); P-101 differential 2 bar (page streamRows).